Animal Nutrition

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Animal Nutrition (A07863)

Short Title:	Animal Nutrition		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Agriculture		
Author	SWALSH		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module will give students a broad introduction to both ruminant and monogastric nutrition for farm animals. Students will learn about the basic principles and key concepts of nutrition, and its importance in diet formulation and prevention of metabolic diseases and disorders.

Programmes

AGRI-0105	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 4 / M

Indicative Content

- Structure and function of ruminant and monogastric digestive systems
- Metabolism of proteins, carbohydrates, fatty acids and amino acids
- Nutrient requirements and intake: macro- and micronutrients
- Systems of energy and protein evaluation of feeds
- Feeding standards for maintenance, growth, reproduction and lactation
- Nutrient sources: grass, conserved forages, root crops, cereals & processed forage
- Diet and ration formulation
- Metabolic diseases and disorders

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain digestion and metabolism in ruminants and monogastrics.
2. Identify the nutrient requirements and intake potential of ruminants in terms of energy, protein, fibre and minerals.
3. Identify a range of feed and forage sources.
4. Calculate diet and ration formulations feeding regimes for monogastric & ruminant production systems.
5. Explain the effect of nutrition on metabolic diseases and disorders for dairy, beef, sheep, pigs & poultry.
6. Use a range of laboratory techniques to classify and calculate the nutritional value of feedstuffs for ruminants.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Guest industry speakers.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,5
Continuous Assessment	50%	
Lab Report	50%	3,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including anatomy and functions of ruminant and monogastric digestive systems, as well as a basic knowledge of macro- and micro-nutrients.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes as well as being able to identify and discuss various nutrient sources.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics including how to predict nutritional requirements of the animal and potential nutritional disorders.

70%-100%: The student will be able to demonstrate a comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Chahal, U.S., P.S. Niranjan and S. Kumar. _Handbook of general animal nutrition_. UK: IBDC, 2008.

Supplementary Material

- D'Mello, J.P.F. _Farm Animal Metabolism and Nutrition_. UK: CABI Publishing, 2000.
- Forbes, J.M. _Voluntary Food Intake and Diet Selection in Farm Animals_ . UK: CABI Publishing, 2007.
- McDonald, P., R.A. Edwards, J.F. Greenhalgh and C.A. Morgan. _Animal Nutrition_. UK: Pearson, 2002.
- McDowell, L.R. _Vitamins in Animal and Human Nutrition_. UK: Wiley-Blackwell, 2008.

Requested Resources

- Room Type: Biology Lab